

Ratio, proportion and percentages - Set 1 - Solutions

Specification M3.1-M3.11.

Answer key

Question	Answer	Topic
1	D	Ratio, proportion and percentages
2	E	Ratio, proportion and percentages
3	B	Ratio, proportion and percentages
4	C	Ratio, proportion and percentages
5	F	Ratio, proportion and percentages
6	D	Ratio, proportion and percentages
7	E	Ratio, proportion and percentages
8	C	Ratio, proportion and percentages
9	C	Ratio, proportion and percentages
10	B	Ratio, proportion and percentages
11	G	Ratio, proportion and percentages
12	B	Ratio, proportion and percentages

Worked solutions

Question 1 RAT-01 · Ratio, proportion and percentages · spec M3.8

The price of a coat is reduced by 20% in a sale, and then the sale price is reduced by a further 25%. The final price is what percentage of the original price?

Answer: D

$$0.8 \times 0.75 = 0.6, \text{ so } 60\%.$$

Question 2 RAT-02 · Ratio, proportion and percentages · spec M3.8

After a 15% increase, a salary is £27 600. What was the original salary?

Answer: E

$$1.15 \times S = 27600 \Rightarrow S = 24000.$$

Question 3 RAT-03 · Ratio, proportion and percentages · spec M3.4

£600 is divided between A , B and C in the ratio $2 : 3 : 7$. How much more does C receive than A ?

Answer: B

Each part is £50; $C - A = 5 \text{ parts} = £250$.

Question 4 RAT-04 · Ratio, proportion and percentages · spec M3.9

y is inversely proportional to the square of x . When $x = 2$, $y = 9$. Find y when $x = 6$.

Answer: C

$$y = \frac{k}{x^2}, k = 36. y = \frac{36}{36} = 1.$$

Question 5 RAT-05 · Ratio, proportion and percentages · spec M3.11

A bacteria population doubles every 3 hours. By what factor does it grow in 24 hours?

Answer: F

$$24/3 = 8 \text{ doublings: } 2^8 = 256.$$

Question 6 RAT-06 · Ratio, proportion and percentages · spec M3.10

Two similar solids have surface areas in the ratio $4 : 9$. Find the ratio of their volumes.

Answer: D

$$\text{Length ratio } 2 : 3, \text{ so volume ratio } 2^3 : 3^3 = 8 : 27.$$

Question 7 RAT-07 · Ratio, proportion and percentages · spec M3.5

A mixture contains sand and cement in the ratio $5 : 2$. How much cement must be added to 35 kg of the mixture to make the ratio $5 : 3$?

Answer: E

35 kg contains 25 kg sand and 10 kg cement. For $5 : 3$ with 25 kg sand, cement must be 15 kg; add 5 kg.

Question 8 RAT-10 · Ratio, proportion and percentages · spec M3.11

£2000 is invested at 5% compound interest per year. Which expression gives the amount after n years?

Answer: C

Compound growth multiplies by 1.05 each year.

Question 9 RAT-13 · Ratio, proportion and percentages · spec M3.8

The value of a car falls by 20% each year. After how many complete years is it first worth less than half its original value?

Answer: C

$0.8^3 = 0.512 > 0.5$, $0.8^4 = 0.4096 < 0.5$. So 4 years.

Question 10 RAT-14 · Ratio, proportion and percentages · spec M3.5

A recipe for 12 biscuits uses 150 g of flour and 100 g of butter. Ann has 1 kg of flour and 500 g of butter. What is the greatest number of biscuits she can make?

Answer: B

Flour allows $\frac{1000}{150} \times 12 = 80$; butter allows $\frac{500}{100} \times 12 = 60$. Butter is the limit: 60.

Question 11 RAT-15 · Ratio, proportion and percentages · spec M3.8

Which of the following statements is/are true?

- I** Increasing a quantity by 50% and then decreasing the result by 50% returns it to its original value.
- II** A 10% increase followed by a 10% increase is a 21% increase overall.
- III** If A is 25% more than B , then B is 20% less than A .

Answer: G

I is false: $1.5 \times 0.5 = 0.75$. II: $1.1^2 = 1.21$, true. III: $A = 1.25B \Rightarrow B = 0.8A$, true.

Question 12 RAT-08 · Ratio, proportion and percentages · spec M3.8

A quantity increases by $x\%$ and then decreases by $x\%$. The overall change is a decrease of 4%. Find x .

Answer: B

$(1 + \frac{x}{100})(1 - \frac{x}{100}) = 1 - \frac{x^2}{10000} = 0.96 \Rightarrow x^2 = 400 \Rightarrow x = 20$.